Athmeeya M Kashyap

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Education

IIIT Hyderabad 2024 - 2029 (Expected)
B.Tech Computer Science + MS Computational Natural Science *Hyderabad, India*

- Coursework: Data Structures & Algorithms, Discrete Mathematics, Linear Algebra, Probability & Random Processes, Design & Analysis of Software Systems, Machine & Data Learning.

CFAL (Junior College) 2022 - 2024
Higher Secondary, Science (PCM) *Mangalore, Karnataka, India*

Achievements

INMOTC Selection (2022): Selected for INMO Training Camp at ICTS-TIFR, Bengaluru(**1 of 22 from Karnataka**).
INMO Qualifier (2022): Qualified for the Indian National Mathematical Olympiad (**1 of 300 students nationally**).
NSEA, Top 1% Karnataka (2023): Top 1% statewide in the National Standard Examination in Astronomy (IAPT).
SGFI National Swimming: Represented Karnataka at the School Games Federation of India Nationals.

Professional Experience

Software Engineer Jan 2026 - April 2026
SVP India (Social Venture Partners India) *Hyderabad, India*

- Engineered a **Partner Engagement Dashboard** for **SVP India**, commissioned by **Mr. Raja Poladi** (Head of CDITH, IIIT Hyderabad), managing **100+ partner and event entries**.
- Designed a normalized relational schema with **12 tables** and **60+ REST API endpoints** using **React.js, TypeScript, Express.js, Prisma ORM, and PostgreSQL**, secured with **JWT/bcrypt** authentication.
- Systematized bulk partner data ingestion via custom seed scripts, eliminating 100% of manual data entry overhead.

Projects

FlowMax: Maximum Flow Algorithm Benchmarking Suite | *Python, NetworkX, Matplotlib* 2025

- Implemented and benchmarked **5 max-flow algorithms** (Edmonds-Karp, Dinic's, Push-Relabel, Hopcroft-Karp, Boykov-Kolmogorov) across **21 datasets** spanning 7 topology categories.
- Empirically confirmed $O(V^2E)$ vs. $O(VE^2)$ complexity gaps with practical case studies in airline scheduling, image segmentation, and optical fibre bandwidth optimization.

You Matter Wellness: Full-Stack Gamification Platform | *React.js, Express.js, Python, Flask* 2025

- **Ranked 9th out of 100+ teams** at the Star Union Dai-ichi Life Insurance National Tech Hackathon; developed a wellness gamification platform with streak tracking, challenge systems, and community leaderboards.
- Architected a **decoupled microservice system** utilizing a React 18 frontend with Framer Motion and Chart.js analytics.
- Built an Express.js REST API layer and a **Python/Flask ML microservice** serving personalized wellness forecasts; deployed on **Vercel** and **Railway**.

Research Experience

Undergraduate Researcher, CCNSB Lab, IIIT Hyderabad 2026 - Present
Complex Networks, Percolation Theory & Dynamical Systems *Hyderabad, India*

- Collaborating with **Prof. Chittaranjan Hens** on **complex network systems**, **percolation theory**, and **dynamical phase transitions**.
- Investigating **signed weighted edge dynamics**, and **structural balance** in mathematical models.

Erdos-Renyi Random Graph Evaluation | *Python, NetworkX, SciPy, Manim*

- Designed and executed large-scale **Monte Carlo simulations** (1,000+ test iterations, up to 10,000 vertices).
- Empirically confirmed the $s^{-3/2}$ cluster-size power law at the **percolation threshold** and benchmarked $O(n^2)$ vs. Batagelj-Brandes linear time generation algorithms.
- Modeled and analyzed dynamical phase transitions across graph networks, culminating in an upcoming research manuscript.

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, C, C++, Kotlin, Bash/Shell
Frameworks & Libraries: React.js, Node.js, Express.js, Prisma ORM, Tailwind CSS, Socket.IO, Jetpack Compose, Room, NumPy, Matplotlib, Pandas, Scikit-learn, Manim
Databases & Tools: PostgreSQL, MongoDB, SQLite, Vite, Git, GitHub, Docker, AWS, CI/CD, Linux
Core Competencies: Algorithm Design & Analysis, Data Structures, Machine Learning, Deep Learning, REST API, System Design, Distributed Systems, Object-Oriented Programming, Unit Testing, Graph Theory & Network Analysis, Statistical Analysis, Probability Theory, Full-Stack & Android Development, Scientific Computing